

Getting to know your Java Programming Environment



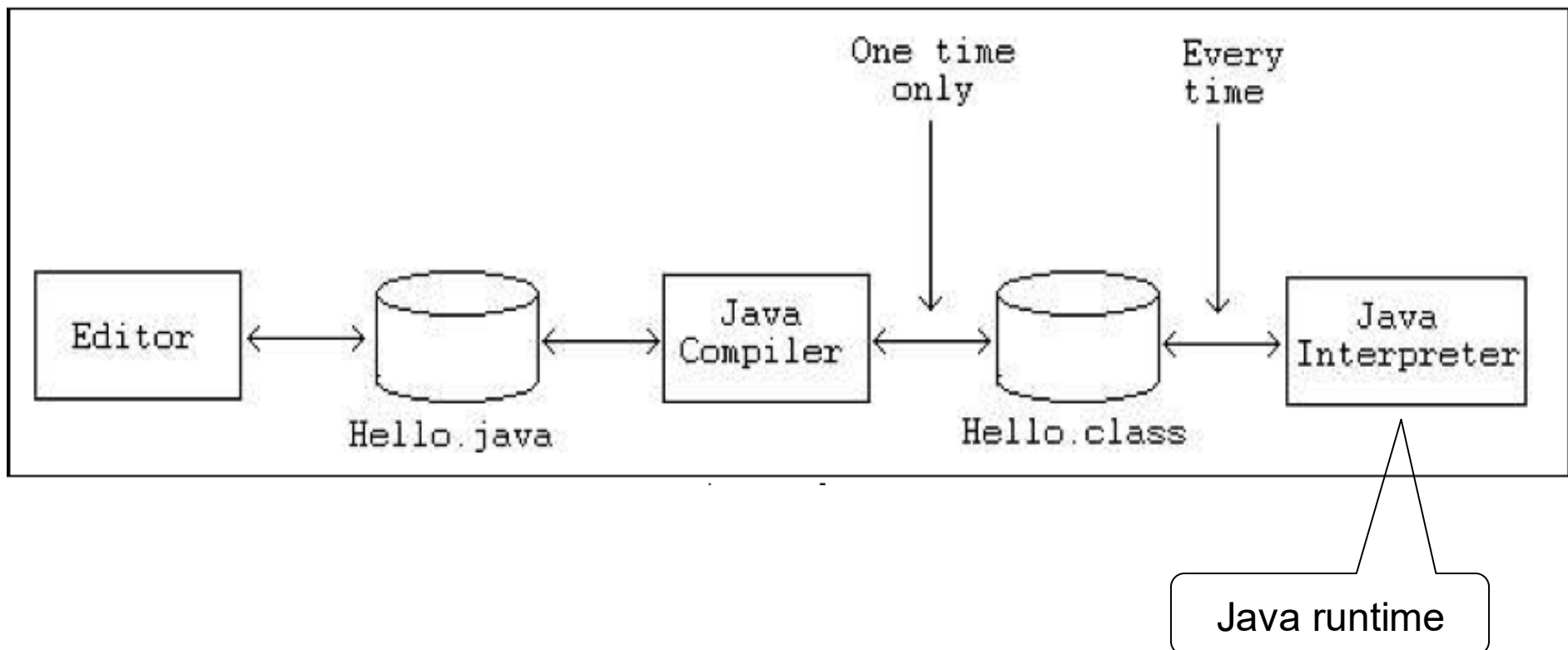
Topics

- Java Program Development
- Build and run a Java program using IDE (IntelliJ/VSC/Eclipse)
- Build and run a Java program using editor and JDK command line tools
- Differentiate between syntax-errors and runtime errors

Java Program Development

Java Program Development Steps

- The following figure describes the process of compiling and executing a Java program



Java Program Development Steps

<i>Task</i>	<i>Tool to use</i>	<i>Output</i>
Write the program	Any text editor	File with .java extension
Compile the program	Java Compiler	File with .class extension (Java bytecodes)
Run the program	Java Interpreter	Program Output

Java Development Environment

- Option #1 – Use your editor of your choice for coding and then use command line Java tools from JDK such as “javac” (compiler) for compiling and “java” (Java runtime) for execution
- Option #2 – Use IDE (Integrated Development Environment)
 - > IntelliJ IDEA
 - > VSC
- In general, IDE provides much more pleasant and productive development environment, so all Java developers use IDE of his/her choice for Java programming

Build and run Java Program using Command-line Tools

Tools to Use

- Text Editor
 - > Examples: Sublime Text, Textmate (Mac only), Notepad (Windows only), Vi or VIM, Emacs
- Java command line tools – are available <JDK>/bin directory
 - > *javac* – compiler, used for compilation
 - > *java* – java runtime (java interpreter), used for execution

My First “Helloworld” Java Program

```
1  public class Hello {  
2      /**  
3       * My first Java program  
4       */  
5      public static void main( String[] args ){  
6          //prints the string “Hello world” on screen  
7          System.out.println(“Hello world”);  
8      }  
9  }
```

Use Text Editor & Command-line Tools

- Step 1: Using text editor, write Java program
 - > Name it "Hello.java"
- Step 2: Compile the Java program
 - > *javac Hello.java*
- Step 3: Run the Java program
 - > *java Hello*

Build and run Java Program using an IDE

What is IDE?

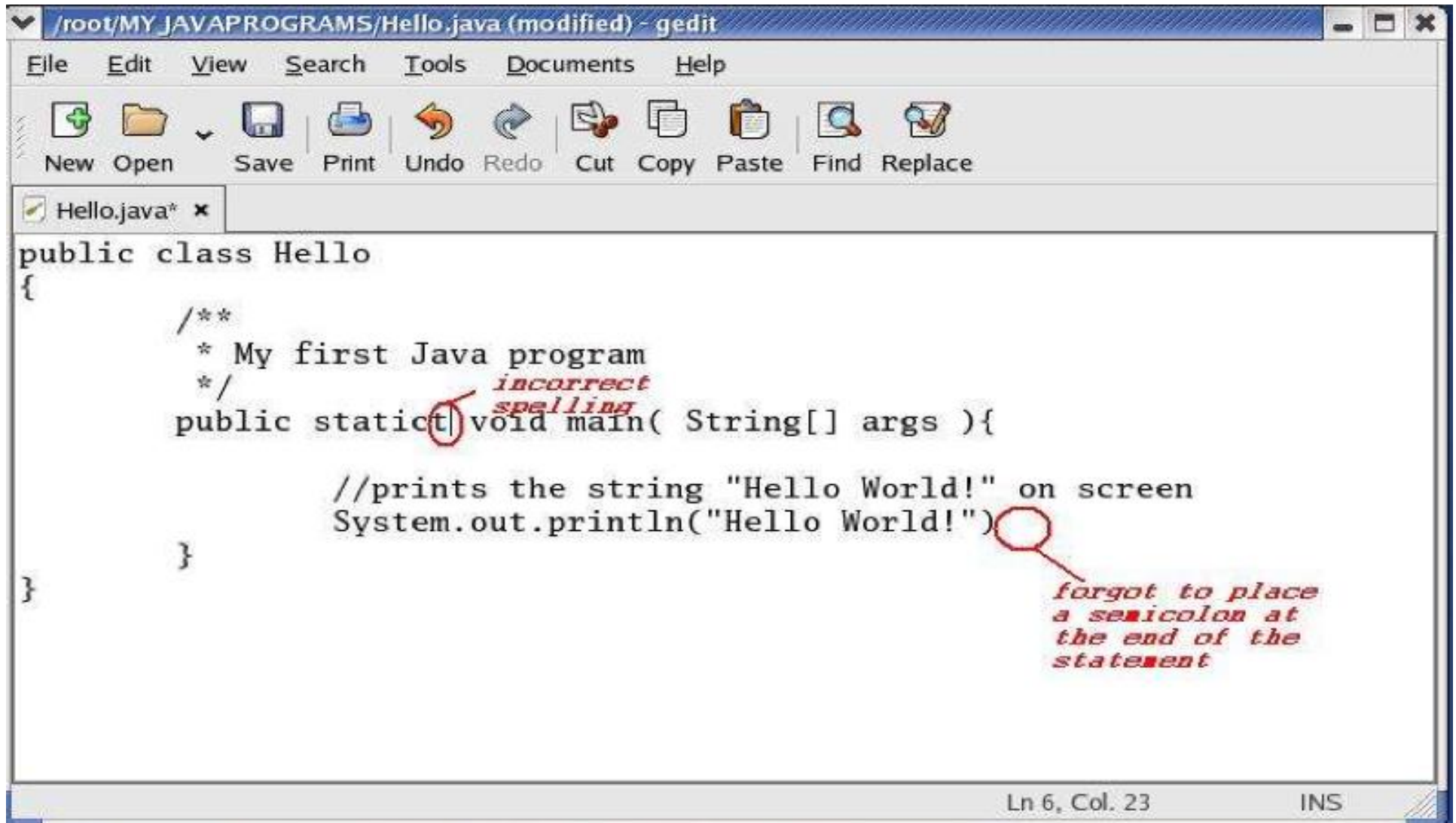
- Integrated Development Environment (IDE) provides complete application development environment
 - > Code editor (with many conveniences such as syntax highlighting)
 - > GUI builder (for building GUI application)
 - > Built-in compiler
 - > Built-in debugger
 - > Built-in tester
 - > Built-in profiler (for performance measurement)
 - > Many more built-in features
- Popular Java IDE's
 - > IntelliJ IDEA (Free/Commercial)
 - > VSC (Free)

Errors: Syntax Errors & Runtime Errors

Errors : Syntax (Compile) Errors

- Syntax Errors
 - > Errors are usually from typing errors
 - > Detected at compile time, hence the reason they are called compile errors
- Common Syntax Errors:
 - > Misspelled a keyword in Java
 - > Forgot to write a semi-colon at the end of a statement

Example: Syntax Error



The screenshot shows a gedit editor window titled "/root/MY_JAVAPROGRAMS/Hello.java (modified) - gedit". The menu bar includes File, Edit, View, Search, Tools, Documents, and Help. The toolbar contains icons for New, Open, Save, Print, Undo, Redo, Cut, Copy, Paste, Find, and Replace. The editor window displays the following Java code:

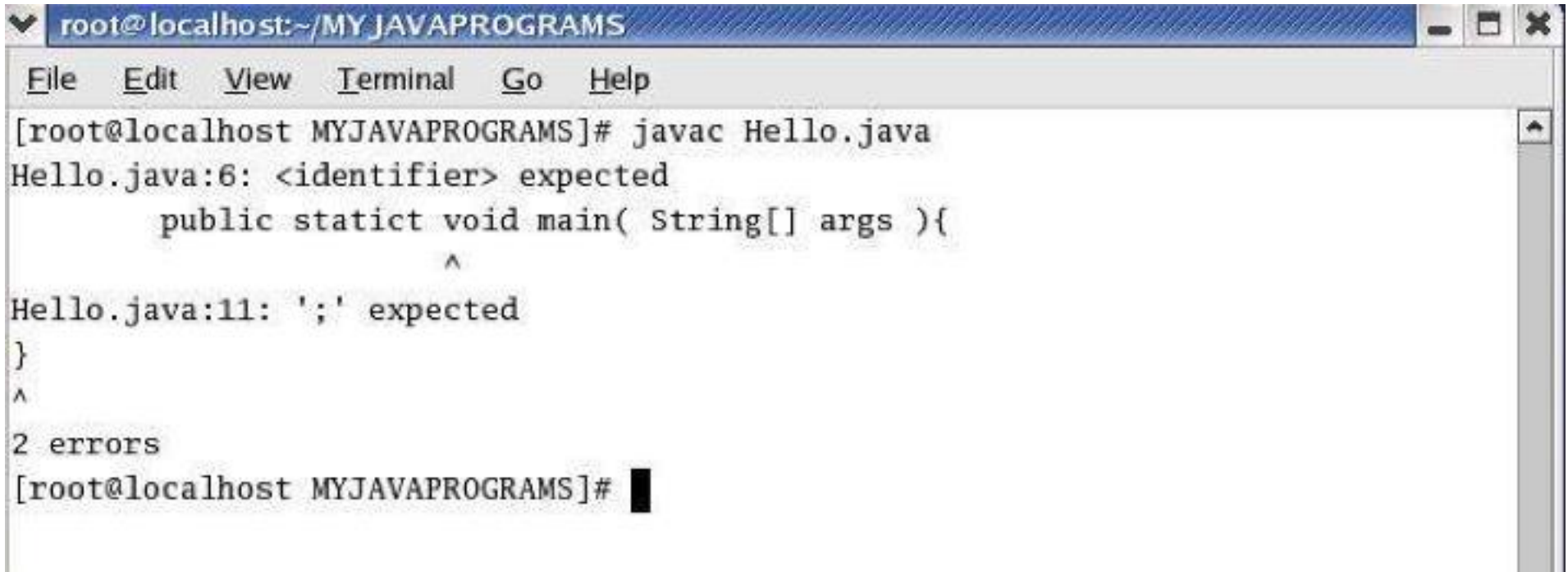
```
public class Hello
{
    /**
     * My first Java program
     */
    public static void main( String[] args ){
        //prints the string "Hello World!" on screen
        System.out.println("Hello World!")
    }
}
```

Two syntax errors are highlighted with red circles and annotations:

- The word `static` is circled, with a red arrow pointing to it and the text *incorrect spelling*.
- The closing parenthesis of the `println` statement is circled, with a red arrow pointing to it and the text *forgot to place a semicolon at the end of the statement*.

The status bar at the bottom right shows "Ln 6, Col. 23" and "INS".

Syntax Error detected by Compiler

A terminal window titled 'root@localhost:~/MYJAVAPROGRAMS' with a menu bar (File, Edit, View, Terminal, Go, Help). The terminal shows the command 'javac Hello.java' and two syntax errors. The first error is 'Hello.java:6: <identifier> expected' pointing to the word 'statict' in 'public statict void main'. The second error is 'Hello.java:11: ';' expected' pointing to the closing brace of the main method. The terminal concludes with '2 errors' and a prompt for further input.

```
root@localhost:~/MYJAVAPROGRAMS
File Edit View Terminal Go Help
[root@localhost MYJAVAPROGRAMS]# javac Hello.java
Hello.java:6: <identifier> expected
        public statict void main( String[] args ){
                ^
Hello.java:11: ';' expected
    }
    ^
2 errors
[root@localhost MYJAVAPROGRAMS]#
```


Errors: Runtime Errors

- Run-time Errors
 - > Errors that will not display until you run your program
- Examples:
 - > You want your program to print 100 strings of “Hello world”, but it only printed 99.
 - > Your program expects a number input from the user, but the user inputted a string, and so your program causes an exception

Thank you!

